RICKY ZHANG

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Education

University of Florida **Gainesville, FL**
Bachelor of Science, Computer Science and Mathematics Aug 2023 – May 2027
Coursework: DSA, OS, Competitive Programming, Combinatorics, Graph Theory GPA: 3.97

Skills

Programming Languages: C++, Java, Python, Lua, Typst, TypeScript, Rust, Bash, SQL
Libraries/Tools: Linux, GoogleTest, Google Benchmark, Boost, FuzzTest, perf, ImGui, Docker, Kafka

Work Experience

Trading Systems Engineering Intern **Jun 2026 – Aug 2026**
Susquehanna International Group (SIG) *Bala Cynwyd, PA*
- Co-built a **kernel-bypass** market maker for 13 equities in a team of 3, reaching **sub-5us p99** tick-to-trade
- Explored "leapfrogging" bracket update dynamics, optimized order book internals, and engineered risk checks
- Designed and benchmarked an **options exchange** from scratch, following a custom wire spec
- Parameterized synthetic order flow and benchmarked pooled vs inline payloads for queue and book entries

Software Engineering Intern **Jun 2025 – Aug 2025**
Sumitomo Mitsui Banking Corporation *New York, NY*
- Built and integrated a **cash-flow** pipeline for interest rate swaps with Spring Boot, Kafka, and RHEL
- Contributed to migration of quant library components (futures, yield curves, swaps) from C++ to Java
- Achieved 30% faster Maven builds for pricer service and QA API by cleaning legacy technical debt

Undergraduate Researcher **Jan 2024 – Aug 2024**
Geoplexity Research Group *Gainesville, FL*
- Devised C++/Nauty algorithms with Dr. Sitharam's group to detect rigid partial 3-tree elements
- Optimized constraint-graph roadmap computation for 60% speed improvements over hash-map methods

Teaching Assistant **Jan 2024 – May 2026**
University of Florida *Gainesville, FL*
- Taught Competitive Programming, C++ Fundamentals, and Discrete Math

Involvement

President — UF Quant Club **January 2026 – Present**
- Lead a **100+ member** quantitative finance organization backed by $3,000 in firm sponsorships
- Hosted weekly interview prep sessions for trading and development roles

Webmaster — UF Society of Asian Scientists and Engineers (SASE) **Jun 2024 – May 2025**
- Hosted workshops for **60+ attendees** on websockets, SSG/SSR architecture, and industry practices
- Spearheaded a team of 18 to develop a website with React Query, TanStack, Tailwind, and Turso

Projects

Vimficiency | *C++, Lua* **Dec 2025 – Present**
- Analyzes editing actions as a **Neovim plugin** that recommends more efficient Vim command sequences
- Uses **Anytime Repairing A*** search across motion/edit/composition optimizers with customizable effort
- Integrates a Lua editor layer with an in-process C++ analysis engine via a C ABI
- Built through **LLM-assisted development** at scale, gated by property, fuzz, and approval test suites

3D Game Engine (Battlebeyz) | *C++, OpenGL, ImGui* **May 2024 – Feb 2025**
- Developed a **3D renderer** and **custom physics engine** from scratch using low-level graphics libraries
- Improved heuristic-based simulation accuracy for **spinning tops** over existing physics engines
- Optimized architecture with multithreading, memory management, and performance profiling

Awards/Certifications

Top 750 Putnam, Akuna Options 201, 3rd Place ICPC Regionals